Extraction of Iron from Hematite

Exam code: 0620 & 0971

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Extraction of iron from hematite

- Iron is extracted in a large container called a blast furnace from its ore, hematite
- Modern blast furnaces produce approximately 10,000 tonnes of iron per day

The blast furnace

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

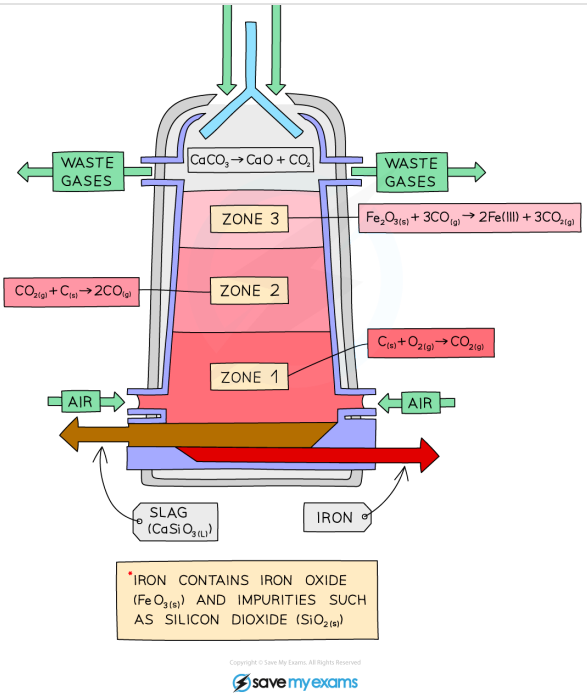

Diagram showing the carbon extraction of iron

- The raw materials: iron ore (hematite), coke (an impure form of carbon), and limestone are added into the top of the blast furnace
- Hot air is blown into the bottom

Zone 1

- Coke **burns** in the hot air to form **carbon dioxide**
- The reaction is exothermic and provides **heat** for the furnace

carbon + oxygen → carbon dioxide

Zone 2

- At high temperatures, carbon (coke) reacts with carbon dioxide to form carbon monoxide.
- Carbon monoxide is the main reducing agent in the furnace.

carbon + carbon dioxide → carbon monoxide

Zone 3

- Carbon monoxide **reduces** the iron(III) oxide to iron
  - Iron is extracted by reduction because oxygen is removed from iron(III) oxide

- The molten iron collects at the bottom of the furnace and is tapped off

iron(III) oxide + carbon monoxide → iron + carbon dioxide

- Limestone (calcium carbonate) is added to the furnace to remove impurities in the ore

calcium carbonate → calcium oxide + carbon dioxide

- The calcium oxide formed reacts with the silicon dioxide, which is an impurity in the iron ore, to form calcium silicate
- This melts and collects as a molten **slag** floating on top of the molten iron, which is tapped off separately

calcium oxide + silicon dioxide → calcium silicate

Examiner Tips and Tricks

For Core students, the symbol equations are **not** needed for the different reactions involved in the extraction of iron from hematite.

Equations for extraction of iron from hematite

Extended tier only

Zone 1

- The burning of carbon (coke) to provide heat and produce carbon dioxide:

$$C\,(s) + O_2\,(g) \rightarrow CO_2\,(g)$$

Zone 2

- The reduction of carbon dioxide to carbon monoxide:

$$CO_2\,(g) + C\,(s) \rightarrow 2CO\,(g)$$

Zone 3

- The reduction of iron(III) oxide by carbon monoxide:

$$Fe_2O_3\,(s) + 3CO\,(g) \rightarrow 2Fe\,(s) + 3CO_2\,(g)$$

- The thermal decomposition of calcium carbonate (limestone) to produce calcium oxide:

$$CaCO_3\,(s) \rightarrow CaO\,(s) + CO_2\,(g)$$

- The formation of slag:

$$CaO\,(s) + SiO_2\,(s) \rightarrow CaSiO_3\,(l)$$

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